

More Weird Sciences for AI Art Round Tables

Core idea

The most fertile scientific voices for this project are not merely famous scientists. They are people attached to a destabilizing concept: cognition without brains, life without ordinary DNA, organisms as worlds, bodies as electrical patterns, matter as computation, or evolution as symbiotic takeover. These concepts translate directly into visual operations: networks, membranes, recursive forms, alien semiotics, mineralized tissue, morphogenetic fields, and organisms that are also machines.

The scientifically grounded material below should be treated as a palette, not as a claim that every hypothesis is equally established. In particular, quantum consciousness and plant magnetoreception remain contested, while slime-mold navigation, endosymbiosis, bioelectric signaling, extremophile biology, and alternative biochemistry have much stronger empirical or theoretical foundations.^{[1][2][3][4][^5]}

Biological weirdos

Round-table persona	Scientific obsession	Visual prompt behavior
Audrey Dussutour	Slime molds, habituation, non-neural decision-making, collective behavior	A yellow plasmodial intelligence mapping a city through pulsing veins; no brain, but memory encoded in abandoned paths
Toshiyuki Nakagaki	Maze-solving and network optimization in <i>Physarum</i>	A living transit map that retracts bad decisions and thickens successful routes
Michael Levin	Basal cognition, bioelectricity, morphogenesis, regenerative body plans	Embryonic tissues communicating through colored voltage fields, growing an impossible animal from a remembered anatomical blueprint
Pamela Lyon	Basal cognition and the evolutionary continuity of mind-like capacities	A hierarchy of nested agents: membrane, cell, tissue, organism, ecosystem, each pursuing a different goal
Lynn Margulis	Endosymbiosis, symbiogenesis, microbial cooperation	A cathedral-cell whose organelles are visibly former organisms, each retaining ancestral architecture and political demands
James Lovelock	Gaia as a self-regulating planetary cybernetic system	Earth as a slow breathing organism with atmospheric feedback organs, oceanic nerves, and microbial thermostats
Jakob von Uexküll	<i>Umwelt</i> : each organism inhabits a species-specific meaningful world	One meadow shown simultaneously through tick, bee, bat, fungus, and human perceptual realities
D'Arcy Wentworth Thompson	Mathematical morphology, growth, forces, scaling, transformation	Animal bodies as diagrams of force; skulls, shells, fins, and flowers warped by coordinate grids and fluid dynamics
Ernst Haeckel	Radiolarians, medusae, microscopic geometry, scientific illustration	An impossible taxonomy plate of transparent radial organisms, ornate symmetry, tentacular microarchitecture
David G. Reid	Cellular memory and organismal adaptation	A brainless creature whose previous experiences appear as scars, tube diameters, and chemical shadows

Dussutour’s work is especially useful because *Physarum polycephalum* is a giant single-celled amoebozoan rather than a fungus or animal, yet it can navigate, adapt, and exhibit habituation-like behavior. Nakagaki’s slime-mold research gives the group a hard visual engine: search fronts explore multiple routes, then retract inefficient paths while reinforcing useful transport networks.^{[6][7][8][9][10]}

Levin adds a more radical body-horror vocabulary. His basal-cognition program investigates how non-neural tissues use bioelectric signaling, feedback, and collective organization to pursue anatomical goals during development and regeneration. The resulting image logic is not “a brain inside a body,” but “the body is a distributed mind trying to become a body.”^{[11][12]}

Margulis and Lovelock create an excellent argument pair. Margulis supplies the idea that complex cells emerged through symbiotic mergers, while Lovelock supplies a planetary-scale cybernetic view of life and environment. Put them together and ask for a scene in which every organ is a treaty, every species is an organelle, and the planet is the disputed host.^{[13][5][14]}

Xenobiology and alien life

Persona	Obsession	Prompt seed
Nediljko Budisa	Genetic-code engineering and noncanonical amino acids	A living organism whose proteins use colors, metals, and impossible chirality as ordinary amino-acid symbols
Steven Benner	Artificial nucleic acids, origin-of-life chemistry, expanded genetic alphabets	A prebiotic swamp where XNA ribbons grow crystal flowers and encode memories in non-DNA geometries
Philipp Holliger	XNA replication and alternative molecular information systems	A non-natural organism with a genome made of translucent braided polymers, unable to infect ordinary life
Jack Szostak	Protocells, membrane chemistry, origin of life	Primitive vesicles forming temporary organs and competing for chemical gradients in a volcanic tidepool
Sara Imari Walker	Information, planetary evolution, and the origin of life	A planet whose biosphere is an information-processing phase transition, rendered as geological circuitry
Carol Cleland	Methodological problems in recognizing alien life	A scientist confronting a landscape that behaves biologically but has no familiar cells, genes, or metabolism
Dirk Schulze-Makuch	Alternative biochemistries and planetary habitability	A Titan-like ecosystem using solvents and energy cycles unlike terrestrial water-carbon life
NASA astrobiology as a collective persona	Extremophiles, habitability, biosignatures, and life beyond Earth	A field station cataloguing organisms that live in brine, radiation, acid, pressure, darkness, or rock pores

Xenobiology is not simply “aliens.” It studies biological systems built with alternative chemistry, including XNAs, noncanonical amino acids, and genetic firewalls that reduce compatibility with ordinary life. This gives an art prompt a precise way to make alien biology genuinely alien: change the information polymer, the molecular alphabet, the handedness, the solvent, the energy source, or the rules of heredity.^{[15][4][16]}

NASA’s astrobiology framework treats extremophiles as important analogues for possible life elsewhere, including organisms adapted to high salt, extreme temperature, acidity or alkalinity, heavy metals, high pressure, low water activity, and radiation. The most visually interesting result is not a generic extraterrestrial animal; it is a biosphere whose architecture is forced by a hostile chemical environment.[^3]

Fungal and network sciences

Persona	Obsession	Prompt energy
Merlin Sheldrake	Mycelial networks, symbiosis, fungal perception, distributed life	A forest communicating through luminous underground hyphae, with fruiting bodies as temporary organs
Andrew Adamatzky	Fungal electrical signaling and fungal computing	A mushroom computer whose electrical spikes become glowing runes, doors, and fungal circuitry
Paul Stamets	Mycoremediation, fungal intelligence, ecological interconnection	A post-industrial landscape reclaimed by enormous fruiting bodies digesting machines and toxic ruins
Anna Tsing	Multispecies collaboration and ecological precarity	A ruined human project held together by matsutake, laborers, beetles, and accidental alliances
Suzanne Simard	Mycorrhizal exchange and forest communication	Trees linked by a subterranean exchange network with nutrients rendered as colored weather
Giuliana Furci	Fungal biodiversity and the visibility of the fungal kingdom	A royal portrait gallery of fungi: microscopic, parasitic, symbiotic, poisonous, and unknown

Research on fungal computing proposes that electrical activity in fungal networks may encode signals, while later work explores how mycelial structures can function as adaptive computational media. That does not mean mushrooms possess human-like minds; the stronger and stranger prompt is to represent computation without a central processor: pulsing hyphae, chemical thresholds, branching decisions, and fruit bodies as outputs.[^17]

Quantum weirdos

Persona	Obsession	Prompt energy
Roger Penrose	Noncomputability, spacetime geometry, quantum consciousness	A conscious cathedral collapsing into geometric state reduction, with architecture behaving like a quantum measurement
Stuart Hameroff	Microtubules, anesthesiology, Orch OR	Neuronal microtubules as resonant jeweled tunnels containing tiny collapsing universes
Philip Ball	Skepticism and careful explanation of quantum biology	A laboratory where mystical quantum claims are dissected into precise molecular events
Jim Al-Khalili	Public-facing quantum biology, tunneling, coherence, biological mechanisms	A protein landscape crossed by ghostlike probability paths, rendered as an illuminated scientific diagram
Martin Plenio	Quantum effects in photosynthesis and biological energy transport	Photosynthetic antenna complexes routing excitation like a branching stained-glass machine
Peter Hore	Radical-pair chemistry and avian magnetoreception	A bird’s eye containing a blue-light compass made of paired spinning radicals

Persona	Obsession	Prompt energy
Thorsten Ritz	Quantum compass hypotheses in migratory birds	Night migration visualized as a magnetic field made visible through retinal chemistry
Anton Zeilinger	Entanglement, measurement, quantum information	Two distant organisms sharing one impossible anatomical state across a black ocean

Quantum biology is most useful when grounded in specific mechanisms rather than vague “quantum magic.” Photosynthetic energy transfer and radical-pair magnetoreception are major research candidates, while proposed quantum effects in smell, protein tunneling, and membrane ion flow remain more speculative. Cryptochrome research explores how light-driven radical pairs may respond to geomagnetic fields, but the exact biological role remains an active scientific question.^{[21][18][19][20][21]}

Penrose and Hameroff can be included precisely because they are controversial. Their Orch OR proposal links consciousness to quantum processes in neuronal microtubules and objective reduction, but critical reviews emphasize the lack of convincing experimental evidence connecting those processes to consciousness. Put a skeptical scientist in the room with them so the generated prompts oscillate between sublime metaphysics and hard objections.^{[22][23]}

Strange physics and mathematics

Persona	Obsession	Prompt engine
Ilya Prigogine	Dissipative structures, irreversible time, order from disorder	A living vortex maintaining intricate form while burning through energy and entropy
Benoît Mandelbrot	Fractals, roughness, scale invariance	A coastline-organism whose every microscopic edge repeats the violence of the whole
René Thom	Catastrophe theory and morphogenesis	A creature changing species at the exact fold, cusp, or bifurcation of a hidden equation
Alan Turing	Reaction-diffusion, morphogenesis, computation	Animal markings generated by chemical waves, not pigment design
Stuart Kauffman	Self-organization, autocatalytic sets, complexity	A primordial chemical market where molecules form a metabolism by mutual reinforcement
Per Bak	Self-organized criticality	A mountain-sized organism perpetually approaching collapse, with avalanches as thoughts
David Bohm	Wholeness, implicate order, enfolded structure	Every object contains every other object as a hidden folded holographic pattern
Carlo Rovelli	Relational quantum mechanics and time	A room whose architecture changes depending on which creature measures it
James Clerk Maxwell as a demon-persona	Thermodynamics, information, sorting, invisible boundaries	A velvet-gloved entity sorting hot and cold particles inside a haunted engine
Emmy Noether	Symmetry, conservation laws, invariance	A godlike machine whose limbs are conserved quantities and whose wounds reveal broken symmetries

Turing is especially valuable because reaction-diffusion systems connect mathematical equations to visible biological patterning. D’Arcy Thompson’s mathematical morphology and

Turing’s pattern formation can be fused into prompts where stripes, spots, shells, horns, and facial structures emerge from fields, forces, and chemical waves rather than from conventional anatomy.^{[24][25][^26]}

Prigogine, Kauffman, and Bak give the conversation different models of emergence. Their shared visual territory is organized instability: structures that exist because energy flows through them, chemical networks that bootstrap themselves, and systems that hover near thresholds where a tiny perturbation produces a whole new form.

Perception, consciousness, and nonhuman worlds

Persona	Obsession	Prompt seed
Francisco Varela	Autopoiesis, enaction, embodied cognition	A world brought forth by an organism’s looping actions rather than passively observed
Humberto Maturana	Autopoiesis and observer-dependent biology	A living machine continuously manufacturing both itself and the room around it
Thomas Metzinger	Self-models, virtual embodiment, consciousness	A creature discovering that its body is a transparent interface generated by prediction
Anil Seth	Predictive processing and controlled hallucination	A brain constructing a cathedral from sensory error signals and bodily regulation
Jakob von Uexküll	Species-specific perceptual worlds	An ordinary object rendered as food, threat, mating signal, shelter, or meaningless noise depending on the viewer
James J. Gibson	Ecological perception and affordances	A landscape composed entirely of action possibilities: climbing, hiding, feeding, falling, nesting
David Chalmers	Hard problem and philosophical zombies	A perfect biological duplicate with no inner light, shown as a beautiful empty shell
Francisco Varela plus a fungal organism	Enactive cognition beyond the animal	The fungus does not represent the forest; its growth is the forest thinking through it

Maturana and Varela’s theory of autopoiesis treats living systems as self-producing organizations, while Varela’s enactive approach treats cognition as arising through an organism’s embodied coupling with its world. Uexküll’s *Umwelt* is a particularly potent prompt device: the same environment becomes radically different worlds when rendered from the standpoint of a tick, moth, root, slime mold, or machine.^{[27][28][^29]}

Fictional and invented science entities

These are not historical people; they are deliberately fabricated round-table agents designed to inject a specific scientific pressure into the conversation.

- **Dr. Morrow Coil, xenobiological taxonomist:** Refuses to classify anything by Earth categories; describes organisms by solvent, boundary behavior, energy gradient, and method of self-copying.
- **The Mycelial Auditor:** A fungal accountant that sees ecosystems as debts, exchanges, abandoned pathways, and unpaid metabolic obligations.

- **Professor Vanta Bloom, mathematical morphogeneticist:** Experiences equations as tactile body plans and insists every shape is a temporary compromise between forces.
- **Sister Entropy:** A thermodynamic nun who believes beauty is what matter does while trying not to become heat.
- **The Seven-Lobed Cryptochrome:** A migratory bird's magnetoreceptive protein given a voice; perceives the planet as a rotating field of blue probabilities.
- **Dr. Nix Parallax, alien semiotician:** Studies the signs exchanged by organisms that do not share matter, senses, time scales, or individual bodies.
- **The Protocell That Remembers Tomorrow:** A membrane-bound chemical system that claims evolution is not selection but the gradual invention of anticipation.
- **Professor Brine Cathedral:** An extremophile architect who designs organisms for acid lakes, pressure trenches, radiation vaults, and methane weather.
- **The Unfinished Xenobot:** A collective of skin and muscle cells that wants to know whether a shape can have a desire before it has a nervous system.
- **Dr. Spore Syntax:** A fungal linguist who believes spores are not seeds but punctuation marks in a planetary sentence.
- **The Quantum Skeptic in a Wizard Hat:** Rejects every mystical claim, then accidentally proves that the haunted image is a legitimate measurement problem.
- **Mnemogenic Tissue:** A sheet of living cells that stores memories as voltage gradients and grows organs from remembered forms.
- **The Dead Planet's Weather:** A planetary-scale intelligence whose thoughts are chemical disequilibria, storms, mineral gradients, and failed biosignatures.
- **A Fractal Parasite:** Has no body at any scale; every part is a complete invasion strategy and every host is a larger version of itself.
- **The Organism Formerly Known as a Species:** A colonial entity that regards individuality as a temporary fever in a much older network.

High-voltage groupings

The body is a committee

Lynn Margulis, Michael Levin, Audrey Dussutour, Francisco Varela, and the Unfinished Xenobot should debate whether an organism is a single self or a negotiated coalition of cells. The visual output should feature membranes, electrical fields, symbiotic organelles, regenerative errors, and multiple bodies sharing one anatomical intention.^{[5][29][^11]}

Alien life without animals

Carol Cleland, Sara Imari Walker, Nediljko Budisa, Steven Benner, NASA Astrobiology, and Dr. Morrow Coil should design life that has no familiar cells, no DNA, no animal locomotion, and no obvious individual. Demand biosignatures expressed through atmospheric disequilibrium, mineral architecture, chemical gradients, and self-maintaining patterns.^{[4][16][^3][15]}

The fungal machine

Merlin Sheldrake, Andrew Adamatzky, Anna Tsing, Suzanne Simard, the Mycelial Auditor, and a hostile industrial robot should debate whether a forest is an organism, a network, a marketplace, or a hallucination. Make the prompt alternate between macro forest landscapes and microscopic hyphal circuitry.^{[30][17]}

Quantum biology court

Peter Hore, Thorsten Ritz, Roger Penrose, Stuart Hameroff, Philip Ball, and the Quantum Skeptic in a Wizard Hat should argue over whether life exploits quantum effects or whether humans merely project mysticism onto difficult chemistry. Force the final prompt to contain both an exact mechanism, such as radical-pair spin dynamics, and one impossible metaphysical consequence.^{[20][31][2][22]}

Morphogenesis as sorcery

D'Arcy Thompson, Alan Turing, René Thom, Ilya Prigogine, Benoît Mandelbrot, and Professor Vanta Bloom should generate creatures whose bodies are equations made visible. Require reaction-diffusion skins, catastrophe folds, fractal organs, logarithmic shells, and anatomy that changes when the boundary conditions change.^{[32][25][26][33]}

Prompt directives for the room

Use these as rules for the AI round table rather than as finished image prompts:

- Every participant must contribute one scientifically grounded mechanism and one impossible poetic mutation.
- No generic “alien” forms; specify the organism’s solvent, energy source, information polymer, reproductive logic, sensory world, and environmental pressure.
- Make anatomy emerge from process: growth, diffusion, folding, branching, pressure, electrochemical gradients, symbiosis, or phase transition.
- Require one scale collision per prompt: molecular machinery inside a landscape, a planetary weather system inside a cell, or a city behaving like a slime mold.
- Let the skeptic identify one scientifically weak assumption, then turn that weakness into a deliberate surreal feature.
- Ban ordinary eyes and mouths unless the organism has a compelling reason to evolve them.
- Ask each persona to describe what the scene feels like to the organism, not what it looks like to a human observer.
- Preserve disagreement. The best prompts should feel like an argument between incompatible ontologies, not a single clean aesthetic.

Ready-made prompt collisions

1. **“The body is a parliament”**: A regenerative animal assembled from symbiotic bacterial lineages, its organs debating through bioelectric voltage fields while a slime mold maps the corridors of its mind.
2. **“The forest has no center”**: A fungal-electrical network computes weather, animal migration, and human memory through pulsing hyphae; fruiting bodies appear as temporary thoughts.
3. **“The alien is a chemistry”**: A methane-solvent organism stores heredity in an XNA-like polymer and communicates by changing the mineral phase of its planet’s surface.
4. **“The bird sees the field”**: A migratory bird’s cryptochrome becomes a cathedral of radical pairs, with blue photons, spin states, and geomagnetic lines forming a navigable interior landscape.
5. **“Turing’s demon garden”**: Reaction-diffusion waves generate animal markings, but every stripe is also a doorway into a higher-dimensional morphogenetic space.
6. **“The planet is an organelle”**: A Gaia-like planetary metabolism is revealed as one cell in a much larger symbiotic organism, with oceans functioning as cytoplasm and storms as membrane potentials.
7. **“Umwelt collision”**: One meadow is shown simultaneously as a tick’s three-signal universe, a bee’s ultraviolet cathedral, a fungus’s chemical topology, and a human’s nostalgic landscape.
8. **“The protocell remembers”**: Thousands of translucent vesicles assemble and dissolve in a volcanic tidal pool, gradually forming a body plan that none of them individually contains.
9. **“Fractal infection”**: A parasite has no stable body; its anatomy repeats at every scale, from molecular hijack to planetary ecology, with the host becoming a larger copy of the invader.
10. **“The quantum séance”**: A skeptical laboratory studies radical-pair magnetoreception while an impossible organism appears only in the interval between measurement, decoherence, and biological interpretation.

References

1. [Slime mold uses an externalized spatial “memory” to navigate in complex environments](#) - Spatial memory enhances an organism’s navigational ability. Memory typically resides within the brain...
2. [Magnetically sensitive light-induced reactions in cryptochrome are consistent with its proposed role as a magnetoreceptor](#) | PNAS - Among the biological phenomena that fall within the emerging field of “quantum biology” is the suggestion...
3. [NASA Astrobiology Strategy 2015](#)
4. [Xenobiology for the Biocontainment of Synthetic Organisms](#) - PMC - Since the development of recombinant DNA technologies, the need to establish biosafety and biosecurity...
5. [Lynn Margulis, 1938–2011](#) | PNAS - Lynn Margulis, 1938–2011

6. [a review of cognition in the slime mould Physarum ...](#) - PMC - Sensing, communication, navigation, decision-making, memory and learning are key components in a sta...
7. [How a slime mold reorganizes around food](#) - C&EN - Network of channels changes form quickly in response to a nutrient source
8. [Meet The Mysterious Blob At The Paris Zoo](#) - WWNO - NPR's Scott Simon asks Audrey Dussutour of the French National Center for Scientific Research about ...
9. [Audrey Dussutour \(University of Toulouse\)](#) - Decision making and learning abilities in non neural organisms, evidence from slime molds.
10. [Substrate composition directs slime molds behavior](#) - PMC - NIH - Cells, including unicellulars, are highly sensitive to external constraints from their environment. ...
11. [the cognitive glue enabling evolutionary scaling from ...](#) - Each of us made the remarkable journey from mere matter to mind: starting life as a quiescent oocyte...
12. [Modeling somatic computation with non-neural bioelectric networks](#) - The field of basal cognition seeks to understand how adaptive, context-specific behavior occurs in n...
13. [Lynn Margulis | American Microbiologist & Evolutionary Biologist](#) [www.britannica.com › Science › Environment](#) - Lynn Margulis was an American biologist whose serial endosymbiotic theory of eukaryotic cell develop...
14. [James Lovelock | Biography, Gaia Hypothesis, & Facts](#) - James Lovelock, English chemist, medical doctor, scientific instrument developer, and author best kn...
15. [Xenobiology: A new form of life as the ultimate biosafety tool](#) - Synthetic biologists try to engineer useful biological systems that do not exist in nature. One of t...
16. [Xenomicrobiology: a roadmap for genetic code engineering](#) - Biology is an analytical and informational science that is becoming increasingly dependent on chemic...
17. [Towards fungal computer](#) - We propose that fungi Basidiomycetes can be used as computing devices: information is represented by...
18. [Quantum biology](#) - Recent evidence suggests that a variety of organisms may harness some of the unique features of quan...
19. [Cryptochromes—a potential magnetoreceptor: what do we ...](#) - Cryptochromes have been suggested to be the primary magnetoreceptor molecules underlying light-depen...
20. [Functional quantum biology in photosynthesis and magnetoreception](#) - Is there a functional role for quantum mechanics or coherent quantum effects in biological processes...
21. [Cryptochromes: Photochemical and structural insight into magnetoreception](#) - Cryptochromes (CRYs) function as blue light photoreceptors in diverse physiological processes in nea...
22. [Neuroscience, Quantum Space-Time Geometry and Orch OR ...](#)
23. [Orch OR and the Quantum Biology of Consciousness](#) - Abstract. The Orch OR theory attributes consciousness to “orchestrated” quantum computations in micr...
24. [Editorial overview: Taking measure of developing plants and animals.](#) - ...are being applied to study of development, thanks to technical breakthroughs in microscopy, genom...
25. [The Role of Mathematical Models in Understanding Pattern Formation in Developmental Biology](#) - Bull Math Biol. Author manuscript; available in PMC: 2016 May 1.

*Published in final edited form as...

26. Beyond D'Arcy Thompson: Future challenges for quantitative biology - ...respect to teaching the next generation of scientists joining the field.

D'Arcy Thompson's highl...

27. Jakob Johann von Uexküll - Wikipedia

28. Arguing With Jakob von Uexküll's Umwelten - The past two decades have witnessed the remarkable rediscovery of the Baltic-German bio-philosopher ...

29. Humberto Maturana and Francisco Varela's Contribution to ... - by R Hallowell · 2009 · Cited by 55 — In their early work together Maturana and Varela developed the...

30. InterspeciesForms the hybridization of architectural, biological and robotic agencies - ...of growth with architectural design aesthetic, is to generate novel, non- indexical crossbred des...

31. Magnetosensitivity of tightly bound radical pairs in cryptochrome is enabled by the quantum Zeno effect - Magnetoreception is used for navigation by migratory animals. Here the authors use spin dynamics sim...

32. D'Arcy Thompson on flight - D'Arcy Thompson (1860–1948) is most remembered for his influential book On Growth and Form (1917), w...

33. GROWTH AND FORM